

T41 V12.8 RF Board Assembly Manual

Version 1.00 – July 31, 2026

WJ Schmidt - K9HZ

INTRODUCTION

The new T41 V12.8 RF board combines the T41 QSE and QSD on a single board, placing the local oscillator onboard and keeping RF traces as short as possible for improved RF performance. The QFN devices—the Si5351 oscillator and two PE4312 attenuators—have been moved to premade plug-in modules that install like ICs. This simplifies soldering and assembly compared with earlier RF board versions, while also supporting experimentation without rebuilding the entire board. The new Oscillator Module uses the more capable eight-output Si5351, enabling automated calibration. The RF Attenuator Modules use one or two PE4312 devices to provide either 0–31.5 dB or 0–63 dB of attenuation.

THEORY OF OPERATION

1. Power.

Board power is supplied as 12 VDC through a connector on the right side of the board. The input feeds 5 V and 3.3 V regulators, which supply the rest of the board circuitry. Total current draw is approximately 100 mA.

2. I2C addressing and switching.

The V12 Main board controls the RF board over I2C. An MCP23017 16-bit I/O expander communicates with the central processor through the SCL and SDA lines on pins 7 and 5 of J3, the “RF CONTROL” connector. In software, the T41 V12 RF board assigns the I/O expander these hexadecimal addresses:

Primary RF board = 0x27

Secondary RF board = 0x23

The user can choose any of the eight chip addresses from 0x20 through 0x27 by configuring the solder switches. A shorted switch sets the corresponding bit to “0”; an open switch sets it to “1”. For the primary board, select 0x27.

The Si5351 oscillator is also controlled over I2C and supports two configurable addresses, 0x60 and 0x61, based on the A0 pin state on the QFN package: A0 = 0 selects

0x60, and A0 = 1 selects 0x61. The T41 V12.8 uses 0x61 so the radio can detect the new Oscillator Module and perform automatic calibration. If a second receiver is deployed using another V12.8 board, its oscillator address is 0x60.

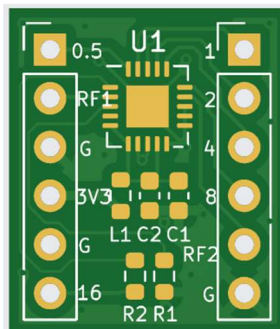
For more information about I2C assignments, see: [T41 V12.6 I2C Assignments](#)

NEW MODULES FOR THE T41 V12.8 RF BOARD

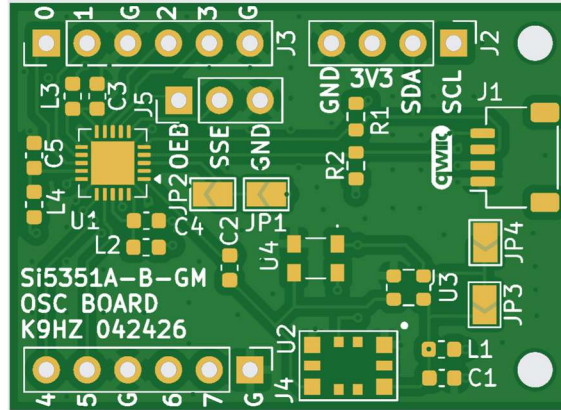
The V12.8 RF board uses three plug-in modules that install like older PDIP 7400-series logic ICs: one Oscillator Module per RF board and Attenuator Modules as needed. Use two Attenuator Modules for an RF board that both receives and transmits, or one for a receive-only secondary board. Because the modules contain QFN components that are difficult to hand-solder without hot-air tools, they are supplied with those components preinstalled. To finish each module, solder on the attachment pins and insert it into the matching board location in the correct orientation. This design simplifies assembly of the V12.8 RF board.

First, cut the required rows of male and female IDC pins for the modules. Machined pins and sockets are preferred because they mate precisely and reduce fit and contact problems. Stamped versions may also be used, but ensure the pins and sockets are compatible because they come in different sizes and shapes, including square and rectangular styles.

For Attenuator Modules U3 and U11, prepare four six-pin rows of pins and sockets—two rows for each module. See below:



For Oscillator Module U12, also prepare two six-pin rows and one four-pin row of pins and sockets:

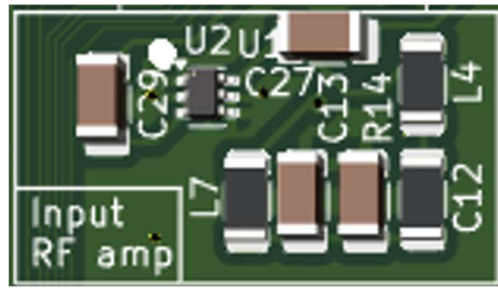


Set the Oscillator Module address at this stage. For a primary receiver, close JP1 with a small solder bridge to set the Si5351 I2C address to 0x61. For a secondary receiver, close only JP2 to set the address to 0x60. Do not close both JP1 and JP2, because that will short the 3.3 V supply rail to ground.

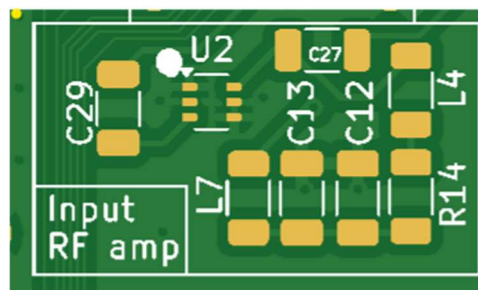
For best alignment, solder the module-board pins and sockets at the end of RF board assembly. First insert the pins into the sockets, then solder the pins and sockets to their boards at the same time.

ASSEMBLING THE T41 V12.8 RF BOARD

1. Before proceeding, verify all parts against the V12.8 RF board BOM on GitHub. Follow the recommended build sequence, because installing components out of order can restrict soldering access. Clean the board with isopropyl alcohol to remove oil and grease before soldering.
2. Install the ICs and other semiconductors first, but leave power regulators U14 and U15 until the final assembly stage. Install U1, U2, U4, U5, U6, U7, U8, U9, U10, U13, U16, U17, U21, U106, and U107 at this stage.
3. Set the RF control I2C address near U21 using jumpers JP15, JP16, and JP17. For the primary RX board, leave all jumpers open to select 0x27. For the secondary RX board, close JP17 to select 0x23.
4. Next, install the capacitors, resistors, and bead inductors. If the V12.8 RF board has the Input RF Amp layout shown below, carefully verify component placement.



Be careful not to swap C12 and R14. The RF board labeling should appear as shown below; this labeling issue was corrected in later board batches.

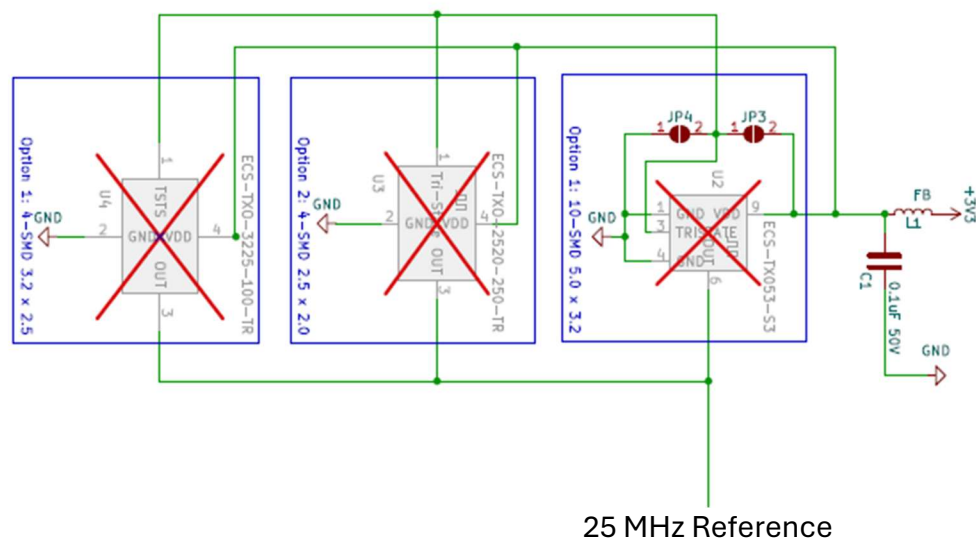


When installing resistors, capacitors, and bead inductors, begin near a previously installed semiconductor and work outward in a circular pattern to preserve soldering access as the PCB fills.

5. Install transformers TR1 and TR2.
6. Install the test point pins, including TP1, TP2, and any remaining test points.
7. Install the remaining parts: voltage regulators U14 and U15, the 10-pin IDC box connector at J3 "RF Control," the Oscillator Module, the Attenuator Modules, and the power connector at J4. To make power connections clear, mark the "+" and "-" screw terminals with red and black dots (with Sharpie pens). Install J1, J5, and J6.
8. After assembly, remove flux with 90% or higher isopropyl alcohol. If needed, scrub the board with Dawn dish detergent and water using a toothbrush, then dry it with a hair dryer on the cool-air setting.

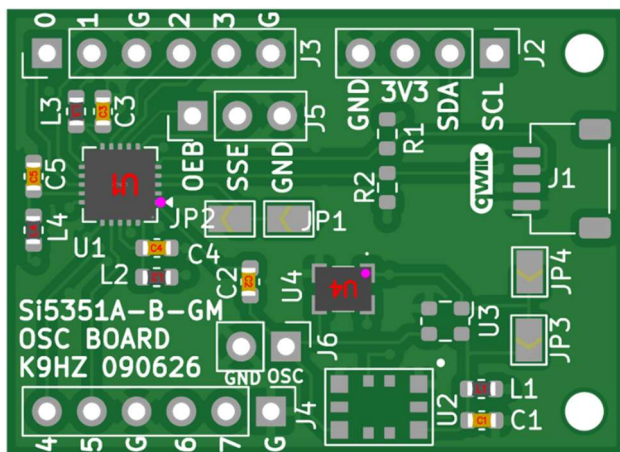
TESTING

To test the completed V12.8 RF board, connect the RF Control cable to the main board and apply 12 V to J4, observing the correct polarity. Then run either the I2C test routine or the Phoenix software on the main board to confirm that it detects the RF board at the correct I2C addresses. No additional testing is performed at this time.



DIVERSITY CONNECTIONS

The latest Oscillator Modules can route the Primary RF board's TCXO signal to the Secondary RF board for Diversity reception. This requires the oscillators to be synchronized using the same TXCO signal in the same phase. The board image below shows the TCXO signal output connection.



Use J6 as the TCXO signal connection point. Remove the TCXO from the Secondary RF board's Oscillator Module, then connect J6 on the Primary RF board's Oscillator Module to the Secondary RF board's Oscillator Module with a jumper. The jumper may be made from IDC pins, IDC connectors, and a short length of small 50-ohm coaxial cable. Any phase difference caused by the coax length is corrected in software during diversity calibration. After calibration, diversity mode can use two antennas to determine the directions of transmitted signals and noise and to null interference.

DUAL PE4312 ATTENUATOR MODULES

Dual Attenuator Modules provide greater attenuation for RF input and output. A single-PE4312 module offers 0–31.5 dB of attenuation in 0.5 dB steps, while a dual-PE4312 module provides 0–63 dB in 1 dB steps. The single module is sufficient for most applications, but experimenters seeking greater attenuation may prefer the dual version. To use it, update the configuration file so the software supports the dual Attenuator Module.

